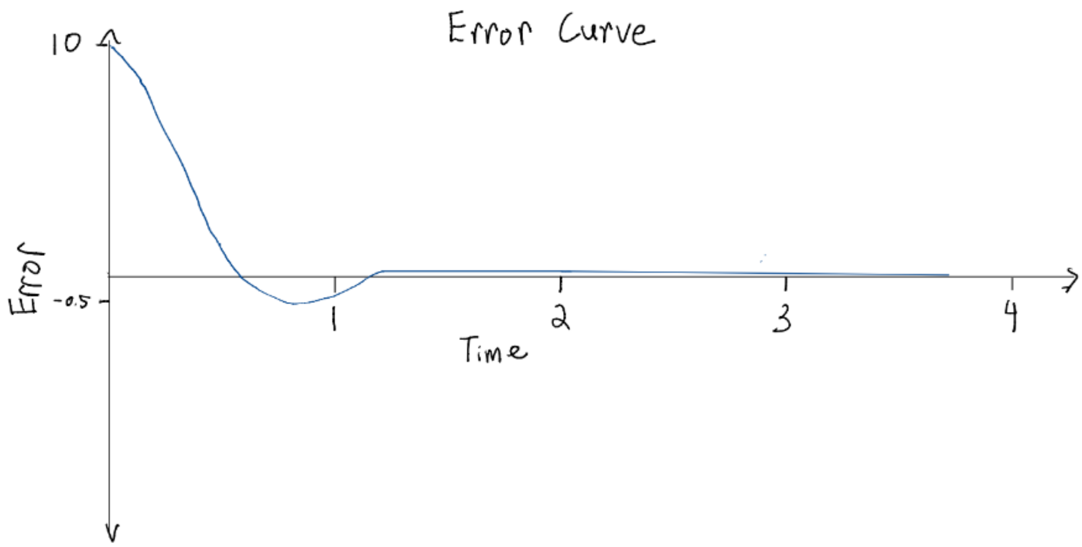
HW

2 PID Tuning

2.2.1 The True Value & Error Curves



2.2.2 Explain an Effect

1. (a) If the absolute value of K_p is too large, there will be a significant overshoot. This causes large rapid oscillations resulting in instability. If the absolute value of K_p is too small, the system will have a very slow response.

   (b) With just P control, it is not possible to tune K_p to eliminate oscillations. This is because there is no damping, so the controller will overshoot and then reverse to correct repeatedly.

   (c) The controller will not stabilize at the setpoint with just P and D control because this model does not account for accumulated systematic errors. For instance, the downward acceleration has a consistent effect on error when flying a drone. With just P and D control, the drone will eventually maintain a steady altitude, but without the I term, this altitude will be lower than the set point. In very special cases where there aren't any outside sources of error, it may be possible to stabilize the set point with just P and D control.

2. (a) The rise time decreases when K_p increases because the control output increases proportionally with K_p. This means that with a higher K_p the system will react more quickly and have a shorter rise time

(b) The settling time increases when K_i increases because the accumulation of error can cause an overshoot. As K_i increases, this overshoot becomes larger and it will take longer for the output to stabilize at the steady state value.

(c) The overshoot decreases when K_d increases because K_d has a damping effect, slowing down the "momentum" of the control output as it approached the steady state value. As K_d increases, this damping effect increases, which lowers the overshoot.

2.2.3 Start Tuning

(a)
1. K_d is too small.
2. Increasing K_d will increase the damping effect and reduce oscillations. The output oscillates around the steady state value, so the K_i term is effective in eliminating steady state error. The system has a reasonable response time, indicating a good value for K_p.

(b)
1. K_d is too small.
2. Increasing K_d will decrease overshoot and decrease settling time. Decreasing K_i will also decrease overshoot and settling time, but since there seems to be little to no steady state error, it is probably better to make more changes to K_d.

(c)
1. K_p is too small.
2. Increasing K_p will help reach the steady state value instead of approaching it asymptotically. Increasing K_i will also reduce the steady state error.

(d)
1. K_p is too large.
2. Reducing K_p will stop the oscillations from growing and create a more stable system. Increasing K_d will increase the damping effect which will make the oscillations converge at the steady state value instead of growing larger.

3 PID on the PiDrone

3.1 Altitude Control

1. The process variable is the altitude. The error is the difference between the desired height of the drone and the measured height of the drone. The control variable is the motor thrust.

2. If the K term is too large, the drone will have a very aggressive takeoff and overshoot the desired altitude. This would contribute to integral windups and large oscillations.

If K is higher than the necessary hovering threshold, the drone may not be able to descend or land safely.

3.2 Velocity Control

1. The setpoint is the desired planar velocity of the drone. The process variable is planar velocity. The error is the difference between the desired planar velocity and the actual measured planar velocity. The control variable is the roll and pitch of the drone.

2. We assume that the velocity PID controller is properly tuned and the drone is hovering in a stationary position prior to 'L' being pressed.

   When 'L' is pressed, the setpoint changes from zero to a constant left velocity. Since the drone cannot change position instantaneously, the error term increases with this jump in setpoint. The drone's roll is adjusted to generate the desired left velocity, which will increase the drone's measured velocity and decrease the error. Note that the roll itself is controlled by a separate PID controller in the flight controller.